

**The Effect of Safranal on the Epithelial-Mesenchymal Transition through
Immunohistochemical Expression of N-cadherin in Squamous Cell Carcinoma Cell
Line**

(In vitro study)

تأثير السفرانال على الانتقال الظهاري الوسيطي من خلال التعبير المناعي للأن-كادهرين في خط
الخلايا السرطانية الحرشفية
(دراسة في المختبر)

Protocol submitted to

Faculty of Dentistry, Cairo University

For partial fulfillment of the requirements for the PhD Degree in Oral and Maxillofacial Pathology

By

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I. Administrative information:

1. Title:

The Effect of Safranal on the Epithelial-Mesenchymal Transition through Immunohistochemical Expression of N-cadherin in Squamous Cell Carcinoma Cell Line (In vitro study).

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5. Roles and responsibilities:

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2. **Omnia Ahmed Badawi** (The Principle Investigator).

II. Introduction:

6a. Statement of the problem:

Squamous cell carcinoma (SCC) of the head and neck is the sixth most common cancer in humans worldwide, nearly more than half a million new cases per year occur. Although treatment improved over the last decades, the five-year survival rate is approximately 50%, which remains one of the lowest among other common cancer types (Jonsson et al., 2012; Jabini et al., 2017).

Despite the important function of chemotherapy in the treatment of cancer, severe side effects are encountered making its clinical application limited. So, it is necessary to search for new anticancer substances with high efficacy and minimal side effects (Jabini et al., 2017). Chemopreventive agents made from natural products may be used as alternative and safe cancer treatment options in comparison to the currently used anti-cancer therapies which are responsible for a wide variety of side effects besides the provided benefits (Surh, 2003; Fryer et al., 2009).

Scientific Rationale

The current study will investigate the effect of Safranal on the biological activities of SCC cell line; namely the epithelial-mesenchymal transition of squamous cell carcinoma cells. This will add to our knowledge about the effect of safranal on SCC, therefore providing a more effective and less toxic possible treatment for this malignancy.

PICO

P: Cultured human SCC cell line.

I: Application of Safranal on SCC cell line.

C: Cultured SCC cell line without application of Safranal.

O: Assessment of epithelial-mesenchymal transition (EMT), migration and cell viability of SCC cell line.

Prioritization of Outcome	Outcome	Measuring Device	Measuring Unit
Primary outcome	Epithelial -mesenchymal transition (N-cadherin immunoexpression)	Image analyzer	area percentage
	Migration (Wound healing assay)	Stereo microscope	μM
Secondary outcome	Cell viability and cytotoxicity (MTT assay)	ELISA reader	% of cells

Research Question:

Does Safranal inhibit epithelial mesenchymal transition of squamous cell carcinoma cell line by suppression of N-Cadherin expression?

6b. Review of literature:

SCC is a common cancer that arises from malignant proliferation of epithelial cells and accounts for a broad range of disease, from superficially invasive to aggressive, highly lethal tumors. It has been noticed that the incidence of SCC has been gradually increasing worldwide over the past several decades (Sham et al., 2016).

Safranal is a monoterpene aldehyde and pharmacologically active component of the saffron (*Crocus sativus*, L.). It is a traditionally used spice and condiment which has been shown to possess potential health benefits and is responsible for the saffron odor and aroma (Winterhalter & Straubinger, 2000; Cragg & Newman, 2005). The antitumor activities of safranal have been reported by many studies, in addition to inhibition of the proliferation and growth by induction of apoptosis in different cancer types such as prostate cancer (Samarghandian et al., 2011; Malaekheh-Nikouei et al., 2013; Samarghandian & Shabestari, 2013). Zhang et al., 2017 reported that Safranal mediates such effects by overcoming epithelial-mesenchymal transition of oral squamous cell carcinoma cells. Still, not much is known about the action of Safranal on the migration and invasion of oral squamous cell carcinoma which requires further investigation.

Cell migration is an essential process during embryonic development, throughout adult life and in some pathological conditions. Cadherins and more specifically the neural cell adhesion molecule N-cadherin, have an important role in migration. N-cadherin is the main molecule during gastrulation and neural crest development in embryogenesis. In cancer, cadherins regulate the balance between suppression and promotion of invasion. E-cadherin acts as an invasion suppressor and is downregulated in most carcinomas, on other hand N-cadherin is usually upregulated and functions as an invasion promote. N-cadherin expression in epithelial cells promotes changes in morphology to a fibroblastic phenotype, making the cells more motile and invasive. (Derycke & Bracke, 2004).

6c. Scientific objectives:

The objective of current study is to investigate the effect of Safranin on the biological activities of SCC cell line; namely the epithelial-mesenchymal transition, migration as well as viability of squamous cell carcinoma cells, therefore providing a more effective and less toxic possible treatment for this malignancy.

III. Methods

A-Samples, intervention and outcomes

7. Calculated sample size

No sample size could be applied on cell line experiments as we generate millions of naturally randomized cells.

8. Description of study sample

Squamous cell carcinoma cell line will be purchased from International Center for Advanced Research (ICTAR-Egypt) supplied with compatible nutrient media.

Inclusion Criteria: Only human cell lines.

Exclusion Criteria: Animal cell lines - Mixed tumors.

Sample preparation:

Cells will be sub-cultured in a 96-well plate.

- 1- Monolayer cells will be incubated for 48 hours at 37°C in the 5% CO₂ incubator.
- 2- After 48 hours, cells will be fixed in wells for 1 hour in trichloro-acetic acid (50%-5 micro liter) which was added to the wells by a multichannel pipette.
- 3- Cells will be then washed with tap water and left to dry for 1 hour.
- 4- Sulphorhodamine B stain will be added (50 µl) and left for half an hour.
- 5- Excess stain will be washed with 5% glacial acetic acid and then attached stain was recovered with EDTA buffer (100 µl).

9. Intervention for each group

- Safranal, it will be purchased from, Sigma Aldrich, Cairo, Egypt.
- Squamous cell carcinoma cell line will be cultured with and without the application of Safranal.

10. Outcomes:

- Expression of N-Cadherin and cell migration (1ry outcomes) and cell viability (2ry outcome).

<i>Prioritization of Outcome</i>	<i>Outcome</i>	<i>Measuring Device</i>	<i>Measuring Unit</i>
Primary outcome	Epithelial –mesenchymal transition (N-cadherin immunoexpression)	Image analyzer	area percentage
	Migration (Wound healing assay)	Stereo microscope	μM
Secondary outcome	Cell viability and cytotoxicity (MTT assay)	ELISA reader	% of cells

-Paraffin embedded cell line pellet preparation:

1. Cell suspension will be spanned at 1800 rpm for 10 minutes at room temperature.
 2. Supernatant will be decanted.
 3. 5 ml of 70% ethanol is added for 30 minutes and vortexed mildly then the same steps are repeated using 100% ethanol until a very solid pellet is obtained.
 4. Finally, using a clean wooden applicator stick, cell pellet will be pulled out carefully of the tube and into a cassette lined with black biopsy filter paper.
- Then Immunohistochemical staining of N-Cadherin will be done according to manufacturer's instructions.

-Wound healing assay:

- 1- SCC cell line will be plated in 24-well plates and grown to confluence.
- 2- The media will be then removed and a scratch will be made using 100 μl pipette tip.
- 3- This will be followed by washing with PBS and treatment with sub-toxic concentrations of safanal for 24 h.
- 4- Wounds will be observed and photographed before (0 h) and after treatment (48 h) using a stereo microscope fitted with digital camera.
- 5- Wound healing will be calculated as the percentage of the initial wound until total wound closure at 48 time point. The assay will be repeated three independent times.

- MTT assay for cellular viability and cytotoxicity:

1. One ml of cells (50,000-100,000 cells/ml) will be plated into each well of 96-well culture

plate for 24 hours before the MTT assay.

2. The cells will be incubated for 24 h in CO₂ incubator.
3. After treatment of cells with antibiotics and antifungal for 24-72 h, experimental media will be removed, and cells will be washed with PBS.
4. Safranin will be added to cell lines, incubated for 48 hours and then will be washed by PBS twice.
5. The cells will be incubated with medium containing 0.5 mg/ml MTT in CO₂ incubator at 37°C for 4 h.
6. The medium will be aspirated, and the formazan product will be solubilized with dimethyl sulfoxide (DMSO).
7. Absorbance at 570 nm will be measured for each well using a microplate reader (BioTek, Flx 800).
8. The percentages of cytotoxicity and cell viability are calculated using the following equations:
$$\% \text{ cytotoxicity} = 1 - [\text{means absorbance of treated cells} / \text{mean absorbance of negative control}], \quad \% \text{ viability} = 100 - \% \text{ cytotoxicity}.$$

B- Assignment to intervention

This study is non-randomized. Randomization is not applied on cell line experiments, as we generate millions of naturally randomized cells, where the amount of cells in each plate will be standardized and each of the 96-well plates will contain the same number of cells (about 25,000-30,000 cells/well).

11. Sequence generation

Not applied on cell line experiments.

12. Allocation concealment

Not applied on cell line experiments.

13. Implementation

Not applied on cell line experiments.

C- Blinding

14. Blinding

Not applied on cell line experiments.

D- Statistical methods

Statistical analysis will be performed with SPSS 16.0 (Statistical Package for Scientific Studies, SPSS, Inc., Chicago, IL, USA) for Windows. Numerical data will be described as mean and standard deviation or median and range, categorical data will be described as number and percentage; Student's t-test will be used to compare between the two groups.

IV- Ethics:

16. Research ethics approval

V-Statement of originality

Safranal can be used as a potential chemopreventive substitute for cancer therapy. Many studies analyzed the effect of Safranal on different types of human cancer cell lines. However, few studies investigate its effect on EMT and migration of SCC cell line. So, this will improve our knowledge about Safranal's effectiveness on such biological activity of this cell line and subsequent ability to use it as a treatment for this type of cancer.

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